

Exam 2 Review: Proof Templates and Practice

Chapter 2

Proof Template – $\lim s_n = \pm\infty$

To show $\lim s_n = +\infty$, using only the definition:

- I. “Fix $M > 0$.” $(M < 0)$
- II. “Let $N = (\text{something})$.” [this requires scratch work]
- III. “Assume $n > N$.”
- IV. Show that $s_n > M$. $(s_n < M)$
- V. “Since $M > 0$ was arbitrary, $\lim s_n = +\infty$.”
 $(M < 0)$ $(\lim s_n = -\infty)$

Activity: Show that $\lim \sqrt{n} = +\infty$. Fix $M > 0$. Let $N = M^2$. Assume $n > N$.

Then $\sqrt{n} > \sqrt{N} = \sqrt{M^2} = M$. Since $M > 0$ was arbitrary, $\lim s_n = +\infty$.

Bonus: Modify the template to show $\lim s_n = -\infty$.

Monotone Sequences

A sequence is **monotone** if it is increasing or decreasing.

Important properties:

- If (s_n) is monotone and bounded, then $\lim s_n$ exists and is a real number.
- If (s_n) is increasing and not bounded, then $\lim s_n = +\infty$.
- If (s_n) is decreasing and not bounded, then $\lim s_n = -\infty$.

Activity: Let $b_n = \sum_{k=1}^n \frac{1}{k2^k}$. Show that (b_n) is bounded and monotone.

Hence (b_n) converges. Monotone: $b_{n+1} - b_n = \sum_{k=1}^{n+1} \frac{1}{k2^k} - \sum_{k=1}^n \frac{1}{k2^k} = \frac{1}{(n+1)2^{n+1}} > 0$

so $b_{n+1} > b_n$ and thus (b_n) is an increasing sequence.

Hint: You may use (as proven on HW2) that $\sum_{k=0}^n r^k = \frac{1-r^{n+1}}{1-r}$ if $r \neq 1$.

bounded: Note that $\frac{1}{k \cdot 2^k} < \frac{1}{2^k}$ for all k , so $\sum_{k=1}^n \frac{1}{k2^k} < \sum_{k=1}^n \frac{1}{2^k} = \frac{1 - (\frac{1}{2})^{n+1}}{1 - \frac{1}{2}} - 1 < \frac{1}{1 - \frac{1}{2}} - 1 = 2$
So (b_n) is bounded above. Since (b_n) is increasing, it is bounded below by b_1 . So (b_n) is bounded. Thus, since (b_n) is bounded and monotone it is convergent.

Subsequences

- **Balzano-Weierstrass:** If (s_n) is bounded, then there exists a convergent subsequence (s_{n_k}) .
- An element $x \in \mathbb{R} \cup \{\pm\infty\}$ is a **subsequential limit** of (s_n) if there exists a subsequence (s_{n_k}) with $\lim_{k \rightarrow \infty} s_{n_k} = x$.

Activity: Construct a sequence where the set of subsequential limits has exactly two elements.

• $((-1)^n) = (-1, 1, -1, 1, \dots)$ $S = \{-1, 1\}$ • $(0, 1, 0, 2, \dots)$ $S = \{0, \infty\}$.

• $(0, 1, 0, 1, 0, 1, \dots)$ $S = \{0, 1\}$ And so many more!

Bonus: Construct a sequence where the set of subsequential limits has exactly k elements.

• $(1, 2, \dots, k, 1, 2, \dots, k, \dots)$ $S = \{1, 2, \dots, k\}$.

Limsup and liminf

Given a sequence (s_n) , let $S_N = \{s_n : n > N\}$. Then

- $u_N = \inf S_N$ and $v_N = \sup S_N$
- $\liminf s_n = \lim u_N$ and $\limsup s_n = \lim v_N$

Important properties:

- $\lim s_n$ exists if and only if $\liminf s_n = \limsup s_n$. In which case,
 $\lim s_n = \liminf s_n = \limsup s_n$.
- $\liminf s_n = \inf S$ and $\limsup s_n = \sup S$ where S is the set of subsequential limits of (s_n) .

↳ Then $-S$ is the set of subsequential limits of $(-s_n)$

Activity: How are $\limsup s_n$ and $\liminf s_n$ related to $\limsup -s_n$ and $\liminf -s_n$

$$\text{so } \liminf (-s_n) = \inf(-S) = -\sup S = -\limsup s_n$$

$$\text{and } \limsup (-s_n) = \sup(-S) = -\inf(S) = -\liminf s_n.$$

Practice

1. Let $(s_n) = (1 + (-1)^n)$. Compute: The set of subsequential limits, $\liminf s_n$, and $\limsup s_n$.

$$S = \{0, 2\}$$

$$\liminf s_n = 0 \quad \limsup s_n = 2$$

2. Let $(t_n) = (1, 2, 1, 3, 1, 4, 1, 5, \dots)$. Compute: The set of subsequential limits, $\liminf t_n$, and $\limsup t_n$.

$$S = \{1, \infty\}$$

$$\liminf t_n = 1 \quad \limsup t_n = \infty.$$

Cauchy

A sequence (s_n) is **Cauchy** if for every $\epsilon > 0$ there exists N such that: if $n, m > N$, then $|s_n - s_m| < \epsilon$.

Important property:

- (s_n) converges (to a real number) if and only if (s_n) is Cauchy.

Proof Template for Cauchy

To show (s_n) is Cauchy, using only the definition:

- I. “Fix $\epsilon > 0$.”
- II. “Let $N = (\text{something})$.” [this requires scratch work]
- III. “Assume $n, m > N$.” If needed, “We can assume $n > m$.”
- IV. Show that $|s_n - s_m| < \epsilon$.
- V. “Since $\epsilon > 0$ was arbitrary, (s_n) is Cauchy.”

Activity: Suppose that $|s_n - s_m| < \frac{1}{mn}$ for all n, m . Show that (s_n) is Cauchy. *let $\epsilon > 0$. Set $N = \frac{1}{\sqrt{\epsilon}}$. Assume $m, n > N$, Then*

$$|s_n - s_m| < \frac{1}{mn} < \frac{1}{N \cdot N} = \frac{1}{(\frac{1}{\sqrt{\epsilon}})(\frac{1}{\sqrt{\epsilon}})} = \epsilon. \text{ Since } \epsilon > 0 \text{ was arbitrary, } (s_n) \text{ is Cauchy.}$$

Bonus: Show that (s_n) is constant.

Chapter 3

Functional Limits via $\epsilon - \delta$

Definition: Let $f: S \rightarrow \mathbb{R}$ and suppose a is the limit of a sequence in $S - \{a\}$.

Then $\lim_{x \rightarrow a} f(x) = L$ if for all $\epsilon > 0$ there exists some $\delta > 0$ such that:

$$\text{if } 0 < |x - a| < \delta \text{ and } x \in S, \text{ then } |f(x) - L| < \epsilon.$$

Theorem ≈ 20.6 : Let $f: S \rightarrow \mathbb{R}$ and suppose a is the limit of a sequence in $S - \{a\}$. Then the following are equivalent:

- $\lim_{x \rightarrow a} f(x) = L$.
- Whenever (x_n) is a sequence in $S - \{a\}$ with $\lim x_n = a$, we have $\lim f(x_n) = L$.

$\epsilon - \delta$ Proof Recipe

To show that $\lim_{x \rightarrow a} f(x) = L$ using the $\epsilon - \delta$ definition:

- I. "Fix $\epsilon > 0$."
- II. "Let $\delta = (\text{something})$." [this requires scratch work]
- III. "Assume $0 < |x - a| < \delta$ and $x \in \text{dom } f$."
- IV. Show that $|f(x) - L| < \epsilon$.
- V. "Since $\epsilon > 0$ was arbitrary, $\lim_{x \rightarrow a} f(x) = L$."

I. "Let (x_n) be a sequence such that $x_n \in \text{dom } f$, $x_n \neq a$ for all n , and $\lim_{n \rightarrow \infty} x_n = a$."

II. Show, using algebraic properties of limits of sequences, that

$$\lim_{n \rightarrow \infty} f(x_n) = L$$

III. Since (x_n) was arbitrary, $\lim_{x \rightarrow a} f(x) = L$.

Activity: Show $\lim_{x \rightarrow a} (-7x + 1) = -7a + 1$.

Let $\epsilon > 0$, set $\delta = \frac{\epsilon}{7}$. Assume $0 < |x - a| < \delta$ and $x \in \text{dom } f$. Then

$$|-7x + 1 - (-7a + 1)| = |-7x + 7a| = |-7(x - a)| = 7|x - a| < 7 \cdot \delta = 7 \cdot \frac{\epsilon}{7} = \epsilon.$$

Bonus: Write a proof recipe to use the sequence definition, then, use the sequence definition to prove the above limit.

Let (x_n) be a sequence such that $x_n \in \text{dom } f$ and $x_n \neq a$, and $\lim_{n \rightarrow \infty} x_n = a$. Then $\lim_{n \rightarrow \infty} (-7x_n + 1) = -7(\lim_{n \rightarrow \infty} x_n) + 1 = -7a + 1$. Since (x_n) was arbitrary, $\lim_{x \rightarrow a} f(x) = L$.